

1. Cloud-Computing

Definition

Cloud Computing provides computing resources over the Internet.

Service Models

- IaaS
- PaaS
- SaaS

Advantages

- Scalability
- Cost Effective
- High Availability

2. Compute-Engine

Google Compute Engine

Features

- Virtual Machines
- Custom Machine Types
- SSD Storage
- Snapshots

Operations

- Create VM
- SSH
- Firewall Rules
- Delete VM

3. Cloud-Storage

Storage Classes

- Standard
- Nearline
- Coldline
- Archive

Uses

- File Backup
- Images
- Videos
- Documents

4. vpc

Virtual Private Cloud (VPC)

Components

- Subnets
- Firewall Rules
- Routes

Benefits

- Secure Networking
- Network Isolation
- Custom Routing

5. IAM

Identity and Access Management

Purpose

Controls user access to Google Cloud resources.

Components

- Users
- Roles
- Permissions

Principle

Grant only the minimum required permissions.